2.0.12

EE25BTECH11012-BEERAM MADHURI

Question:
A product is made by mixing three raw materials 1, 2, 3 in varying proportions, where material 1 must account for not more than 50% of the total. If x, y and z are the amounts of raw materials 1, 2, 3, respectively, this constraint can be modeled as (2007 PI)

a) $x \leq 0.5$
b) $x \leq 0.5(x + y + z)$
c) $0.5x \leq x + y + z$
d) $x \geq 0.5(y + z)$

Solution:
Given,
Raw Materials: 1,2,3
Amount: x,y,z
let,
Total amount of Product: T

$$T = x + y + z \tag{4.1}$$

given constraint: Material one (x) must not account for more than 50% of the total.

$$x \leq 0.5(T) \tag{4.2}$$
$$x \leq 0.5(x + y + z) \tag{4.3}$$
$$x \leq y + z \tag{4.4}$$

Correct Option: b

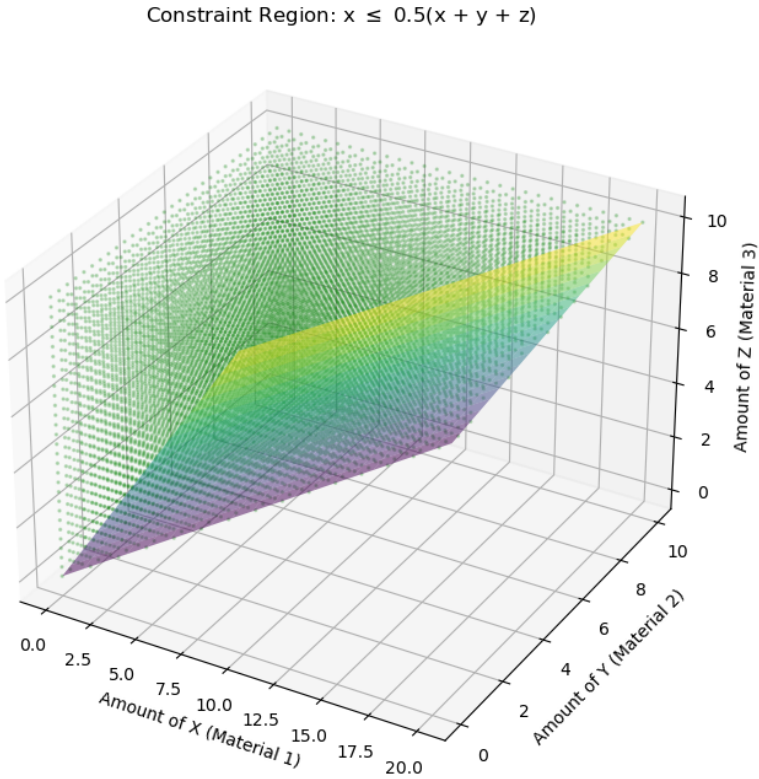

Fig. 4.1: Boundary Condition